CRIMEVISIONAI

Solution Requirements

Folder 2 | Functional and non-functional requirements | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Functional requirements

ID	Requirement	Priority	Current status / acceptance evidence
FR-01	Create, list, view or update case records through the case workflow.	Must	Basic case APIs reported; verify each operation against current build.
FR-02	Upload supported evidence and associate it with a case.	Must	Evidence upload and case-scoped storage reported.
FR-03	Analyze audio and produce a timestamped transcript.	Must	Whisper base pipeline reported; no speaker diarization.
FR-04	Analyze images using object detection, scene similarity and captioning.	Must	RT-DETR, CLIP and BLIP pipeline reported.
FR-05	Analyze supported video by sampling frames and aggregating visual outputs.	Must	1 FPS, maximum 180 seconds / 180 frames reported.
FR-06	Create candidate events / timeline entries from analysis outputs.	Must	LLM-assisted event types reported; validate chronology and source linkage.
FR-07	Allow case-level investigator questions over available analysis.	Should	Case chat endpoint reported; grounding defect known.
FR-08	Return media and analysis results through APIs for the UI.	Must	Evidence/media/analysis routes reported.
FR-09	Export a structured PDF case report.	Should	Not confirmed complete; treat as gap until verified.
FR-10	Support speaker diarization and robust cross-modal identity linking.	Future	Not in known baseline; requires separate models and validation.

2. Non-functional requirements

ID	Requirement	Acceptance approach
NFR-01 Reliability	Errors from malformed uploads, missing models or API failures should be surfaced cleanly.	Test negative paths and record failure messages; not yet represented as a measured pass.
NFR-02 Traceability	Outputs should identify the originating evidence and timestamps where applicable.	Inspect result schema/UI and verify source references.
NFR-03 Performance	Analysis latency should be measured separately for each modality and hardware configuration.	Run repeatable benchmarks; do not use generic <2 s expectations for ML inference.
NFR-04 Security	Protect uploaded evidence and API secrets; do not expose debug mode publicly.	Use local trusted environment; disable debug and add authentication/authorization before deployment.
NFR-05 Maintainability	Keep routes, analysis logic, storage and UI concerns organized.	Review project tree and shared helper functions.
NFR-06 Usability	Show accepted formats, progress, results and actionable errors.	Perform task-based usability walkthrough.
NFR-07 Responsible AI	Avoid definitive claims; distinguish detections from interpretations.	Manual review and Q&A grounding tests.
NFR-08 Portability	Document model downloads, Python dependencies and environment variables.	Clean-environment installation test.

3. Operating constraints

Reported constraints: local CPU with automatic CUDA detection; 100 MB request cap; supported audio formats WAV/MP3/M4A/FLAC/OGG/AAC/WMA; video formats MP4/MOV/AVI/MKV/WEBM; 30-second audio chunks; video maximum 180 seconds at 1 FPS. Availability and speed of local model downloads and Groq access depend on the execution environment.

4. Out of scope and safety requirements

CrimeVisionAI does not determine guilt, replace forensic experts, certify evidence integrity, or provide guaranteed identity recognition. It must not be marketed as court-validated forensic software. Evidence should be synthetic or lawfully authorized during development; any operational use would require security, privacy, legal and forensic-process review.